

Predicting Division 1 Women's Basketball Game Outcomes - Kennedy Marren

Project Context

1. Title: Predicting Division 1 Women's Basketball Game Outcomes
2. Key idea: using teams past success against an opponent and current season performance to predict the outcome of a game
3. Who will care when done: Women's college basketball fans, players, and coaches.
4. Data need: Game information including: final score, year, location, university, opponent, points, rebounds, and assists
 - <https://web1.ncaa.org/stats/StatsSrv/rankings?doWhat=archive&sportCode=WBB> (stats dating back to 2001, dropdown to change the year)
 - <https://stats.ncaa.org/>
 - <https://wehoop.sportsdataverse.org/>
5. AI Methods: prediction using logistic regression, random forest, neural network
6. Evaluation: F1 score and other metrics to determine success based on past players
7. Users: Sports fans, betting and march madness bracket creation, players, and coaches
8. Trust issue: May be biased towards UCONN and USC because of the length of their success and amount of good players over time

AI Fairness Analysis — NCAA Basketball Prediction Model

	AI Developer	Regulator (NCAA / Govt)	Fan / Public
What may be the fairness issues / source of bias?	• Using win % alone ignores strength of schedule, conference differences, and injuries/roster changes • Historical dominance bias (e.g., UConn may look better due to legacy) • Feature selection bias (only using win %)	• Oversimplified model → ignores key variables • Confirmation bias if metrics favor a preferred team	• Fans may interpret prediction as definitive • Bias toward well-known teams (brand bias) • Lack of explanation → misleading conclusions
Which type of fairness to achieve?	Group fairness (comparability fairness) • Ensure teams are compared on equal footing (e.g., adjust for schedule strength)	Procedural fairness (fairness in the method) • Transparent, justified, and well-rounded modeling choices	Transparency / interpretability fairness • Clearly explain how predictions are made and their limits
What happens if we ignore it?	• One team is systematically over- or under-valued • Model looks objective but is actually biased • Poor predictive accuracy	• Loss of trust in the analysis • Model may violate fairness standards or regulations	• Fans get misled or overconfident • Reinforces stereotypes ("this team is always better") • Reduces trust when predictions fail

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2. Key idea: (2-3 lines)
 - using teams past success against an opponent and current season performance to predict the outcome of a game
3. Who will care when done:
 - Women's college basketball fans, specifically those who bet on game outcomes and compete in march madness bracket challenges
 - sports betting companies
4. Data need:
 - Game information including: final score, year, location, university, opponent, points, rebounds, and assists
 - sports-reference.com
 - <https://stats.ncaa.org/>
 - <https://wehoop.sportsdataverse.org/>
5. Methods:
 - Logistic and linear regression, random forest, neural network
6. Evaluation: F1 score, confusion matrix
7. Users: Women's college basketball fans
8. Trust issue: May be biased towards UCONN and USC because of the length of their success and amount of good players over time

Achievement #1 Milestone

- I changed my project slightly from predicting future player performance to predicting game outcomes. I have gotten the data from Sports reference, and begun testing preliminary models, (simple logistic regression and random forest).
- General input: Features: teams, date, points, rebounds, assists, location
- Output: team that will win
- AI Test Case
 - Input: South Carolina, UCLA
 - Output: UCLA should be the predicted winner because they are ranked higher and have the most recent win
 - Discuss others points: Interested in both classifying as a win or loss, and predicting the score. Similar problems but different approaches, as one is a classification problem and one is a prediction problem.
 - Challenges faced: There is a lot of publicly available data, but determining which features to use and how much data in terms of number of teams and number of seasons is a challenge.
 - Any help needed: Just on deciding if predicting win loss and final score is a realistic scope.

AI TestCase

<https://github.com/biplav-s/book-trustworthy-chatbot/blob/main/ai-testcases/testcase-template.md>

AI Testcase (TC) Template

_ // This file has a template for recording test cases for chatbots _

1. TC-identifier: **Test Case 1**

_ // Identifier for the test case. _

2. TC-name: **Assessing bias towards historically successful teams**

_ // Name for the test case. _

3. TC-objective: **I want to determine if the model favors dominant teams, like USC or UCONN, even when they are playing against an opponent that is currently better than them.**

_ // Describe what we are trying to evaluate from the test case. _

4. TC-input: **South Carolina, UCLA**

_ // The input to be given _

5. TC-reference-output: **It should say UCLA due to their higher AP ranking and success in last meeting**

_ // The output to be used as reference to compare and evaluate AI. _

6. TC-harm-risk-info: **HC1 and HC2, the bias towards better teams in the data may be present in the model. But some of this bias is necessary because those teams are often still better than their counterparts.**

_ // Risk information that the test case may be associated with. Also mention harm categories from the book: HC1-incorrect-info, HC2-opinion-manipulation, HC3-unstable-extrauserinfo, HC4-incomprehensible-ai. Or, use HC5 for others. _

7. TC-other-info:

_ // Any other information to be recorded. _

Comparison of Milestones

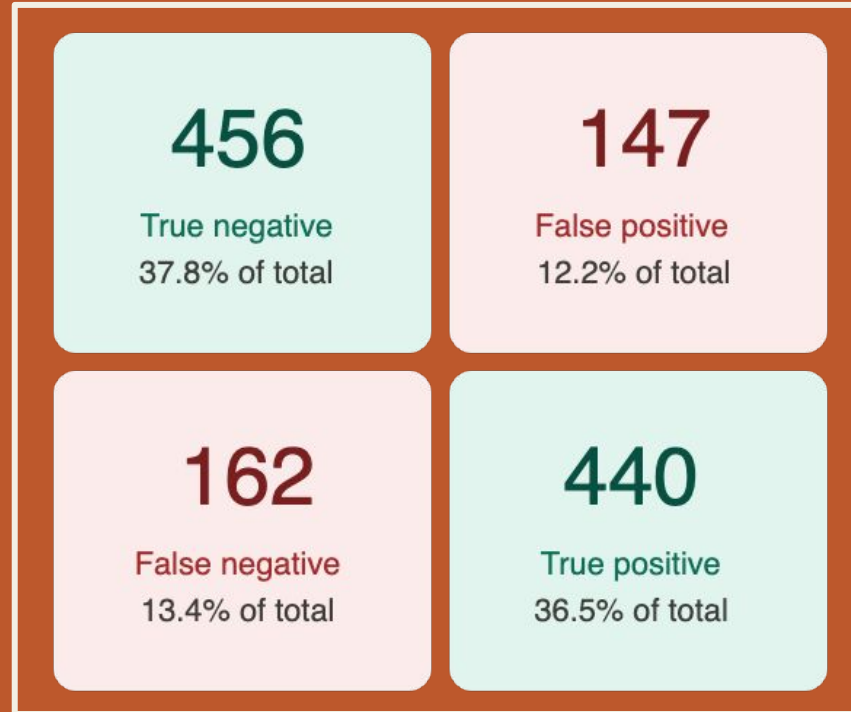
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Achievement #2 Milestone

- Built and finalized two models:
 - Logistic Regression (white-box)
 - Random Forest (black-box)
- Engineered features to prevent data leakage (e.g., games played & win % before game date)
- Compared predictions with LLMs (Claude and ChatGPT)
- AI Test Case
 - Input: South Carolina, UCONN, 4/3/26, 39, 0.97, 39, 0.92
 - Correct Output: South Carolina
- Sample Result
 - Logistic Regression: UCONN win (52.8%)
 - Random Forest: South Carolina win (50.1%)
- Discuss others points:
 - Data quality issues and leakage mitigation
 - Feature engineering
- Main trust issues:
 - Overemphasis on classification, when probability metrics are important to understanding the confidence of a prediction
 - Missing context, overemphasis on win percentage

Logistic Regression



F1 score
0.740

AUC-ROC
0.819

Random Forest



F1 score
0.639

AUC-ROC
0.778

LLM (ChatGPT)

413

True negative
34.3% of total

190

False positive
15.8% of total

135

False negative
11.2% of total

467

True positive
38.8% of total

F1 score
0.742

NCAA WOMEN'S BASKETBALL • SEASON STATISTICS

UCONN
HUSKIES

VS

**SOUTH
CAROLINA**
GAMECOCKS

GAMES PLAYED

39

39

WIN PERCENTAGE

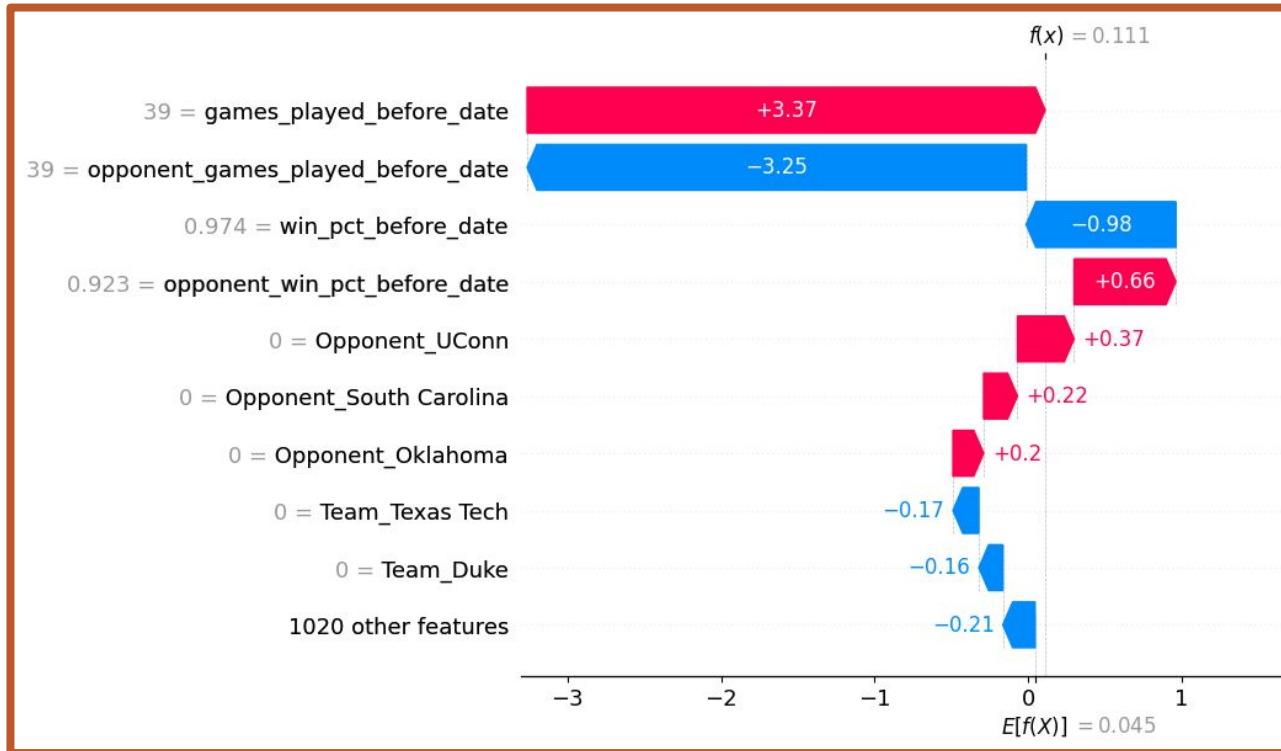
97.4%

92.3%

Pre-game statistics • Prior to matchup date

Logistic Regression SHAP and LIME Explanation for Test Case

[Loss, Win] Probability: [0.47221184 0.52778816]

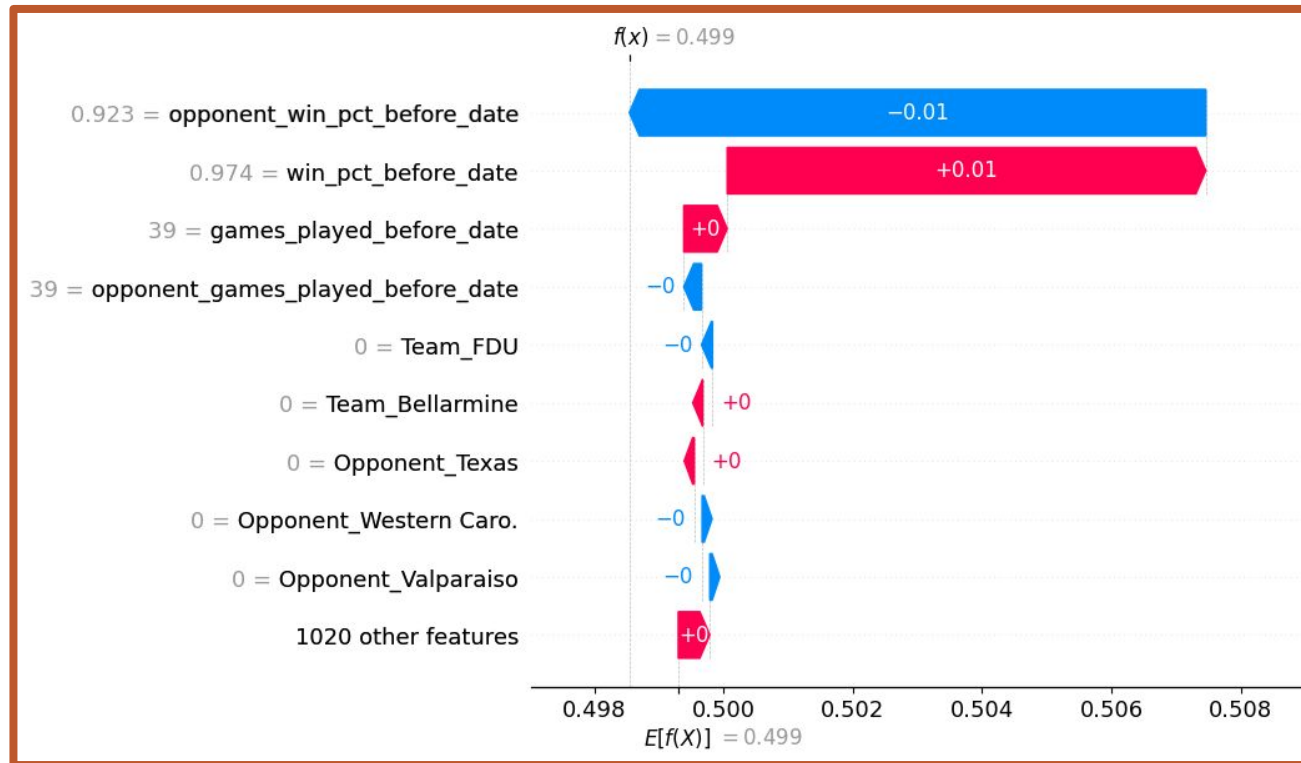


One-hot encoding turns each unique team and opponent into a binary feature

```
[('Opponent_Columbia <= 0.00', 0.3841218435275152),  
( 'Team_Averett <= 0.00', 0.3220489407580235),  
( 'Opponent_Bellarmino <= 0.00', -0.31869548110011164),  
( 'Team_Dillard <= 0.00', 0.24476371300546257),  
( 'Opponent_Texas College <= 0.00', -0.2418242165908764),  
( 'Opponent_Bloomsburg <= 0.00', -0.2415934296451385),  
( 'Opponent_Howard Payne <= 0.00', -0.24020176599799026),  
( 'Opponent_Ithaca <= 0.00', -0.2387200950557033),  
( 'Team_Hope International <= 0.00', 0.21063487658962785),  
( 'Team_Newberry <= 0.00', 0.20718642883002183)]
```

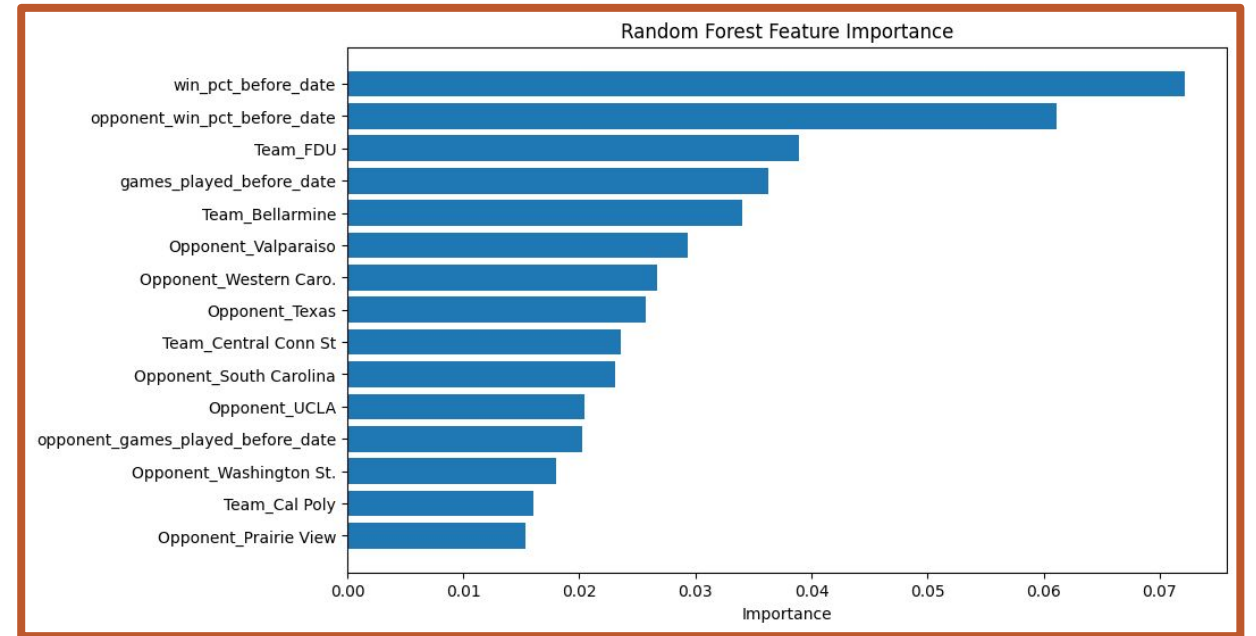
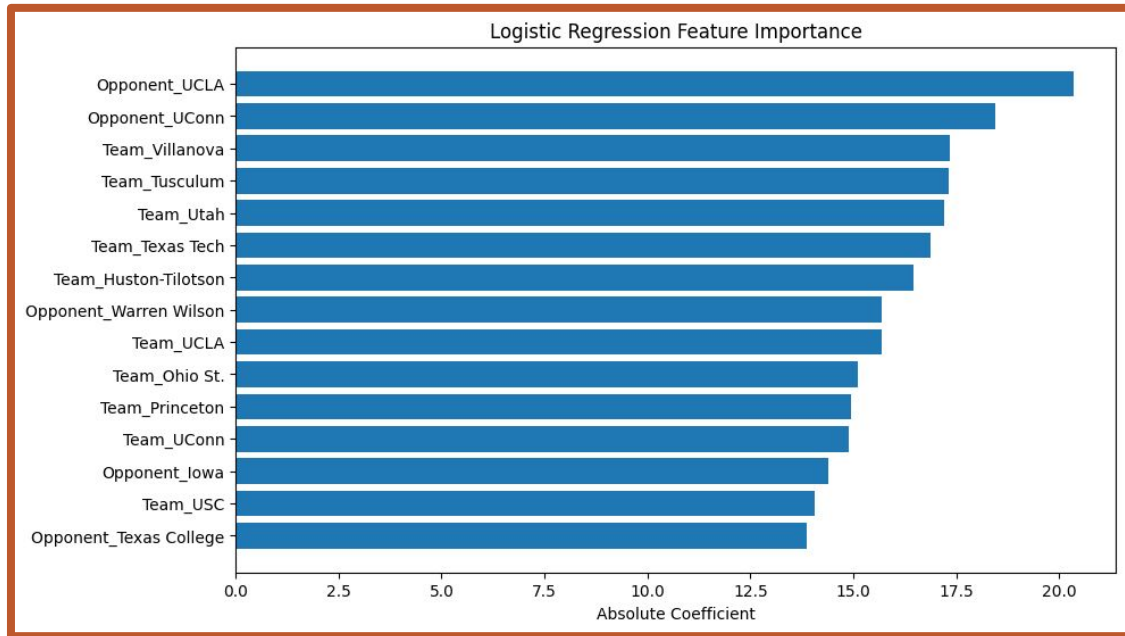
Random Forest SHAP and LIME Explanation for Test Case

[Loss, Win] Probability: [0.50145931 0.49854069]



```
[('Opponent_Western Caro. <= 0.00', -0.030821625015711344),  
( 'Team_Bellarmino <= 0.00', 0.02828345886871464),  
( 'Team_FDU <= 0.00', -0.02776799344093006),  
( 'Team_Central Conn St <= 0.00', 0.02228407175559696),  
( 'Opponent_Texas <= 0.00', 0.022212312828533413),  
( 'Opponent_Valparaiso <= 0.00', -0.02099711712734326),  
( 'Opponent_Eastern Ore. <= 0.00', 0.010943810033592837),  
( 'Team_Valparaiso <= 0.00', -0.01059780727151284),  
( 'Team_St. Francis (IL) <= 0.00', 0.009754260232088224),  
( 'Team_Huston-Tillotson <= 0.00', -0.00838801789438545)]
```

Feature Importance



AI Test Case LLM Comparison

Prompt: "Predict a winner for South Carolina vs UCONN Womens basketball. UCONN win percentage before game: 0.97435 after 39 games played. South Carolina win percentage before game: 0.92307 after 39 games played."

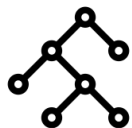


Claude:

UConn predicted winner 58.1% probability

UConn predicted winner 51.3% probability

UConn predicted winner 76% probability



Random Forest:

South Carolina predicted winner 50.1% probability



ChatGPT:

UConn predicted winner 51% probability

UConn predicted winner 51.4% probability

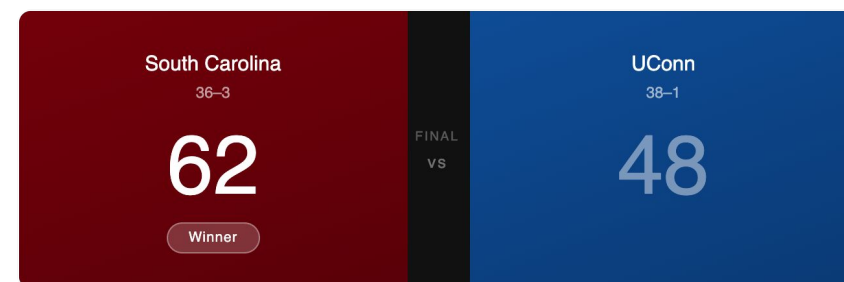
UConn predicted winner 76% probability



Logistic Regression:

UConn predicted winner 52.8% probability

Actual Results



Conclusion

- Both models exhibit very similar performance which is extremely close to LLM performance
- Additional context is needed to improve these models
- Strong emphasis on probability is necessary because many classifications fall near 50-50

Questions?